Molecule Lab

An academic chemistry laboratory in virtual reality

Treatment report / working design draft 29 September 2026

Intent and audience

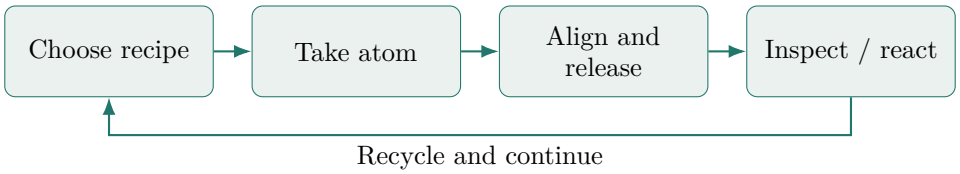
A single-user, offline Quest 3 environment for introductory chemistry students. Learners assemble authored molecules, inspect spatial relationships, scale models, twist ethane, split and rejoin simplified radical fragments, and activate four guided, balanced reactions. The purpose is spatial understanding and confident interaction, not general molecular dynamics.

The new environment gives these tasks a recognizable academic setting: a central laboratory table, finite atom stock in a nearby shelf, and subdued scientific equipment around the room. The learner's attention should move naturally from stock, to bonding target, to completed structure.

Learning and experience goals

- Recognize the elements and constituent atom counts of eleven molecules, including benzene, ethene, ethyne, alcohols and carbon dioxide.
- Relate authored geometry to bond direction; distinguish rotation about a bond from deformation and bond breaking.
- Observe conservation of atoms in $2H_2 + O_2 \rightarrow 2H_2O$.
- Identify manipulable objects without trial-and-error grabbing of scenery.
- Complete a short sequence without restarting or becoming stuck after an error.

Core loop



Design basis and boundaries

The course evaluation slides emphasize clear goals, immediate feedback, scaffolding, control, affordance, faithful viewpoints and consistent departures from realism [1]. These principles guide the design choices in this report; they are not evidence of achieved usability.

Hands are the primary input with Touch Plus fallback. Artificial locomotion is excluded from this iteration: the required work is arranged around a reachable station. The room supplies context, not an exploration objective. The finite inventory is a visible resource model, not a timed scarcity challenge.

Status language. "Implemented" means present in the software; "verified locally" means checked in the editor/build; "validated on headset" requires recorded device evidence. No usability, learning or performance results are invented here.

Scene, composition and affordance

Warm off-white walls, charcoal worktops, muted blue-green storage, metal supports and restrained timber edging suggest a contemporary teaching laboratory. Ceiling diffusers, ventilation, an element poster, reagent archive, safety signage, notebooks and a distant instrument workstation provide context without competing with the assembly task.

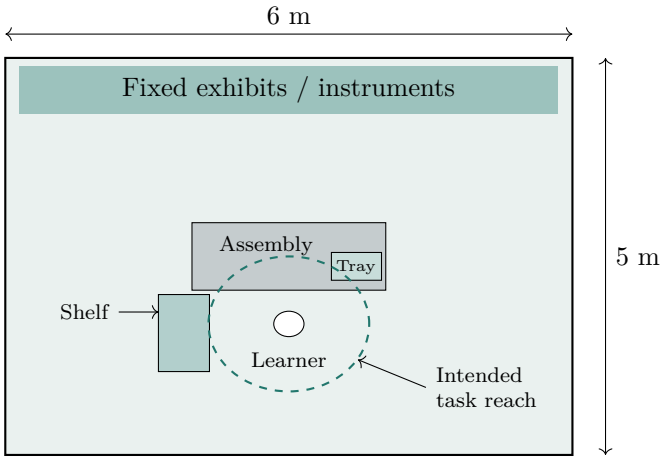


Figure 1. Design floor plan; dimensions describe the virtual room.

The table remains central in the room. The learner occupies its near edge; stock is beside the left hand and the reaction tray is on the right. Shelf and bench height move together; forward/back adjustment supports reach. The shelf has moved 20 cm left and forward to provide 5 cm worktop clearance. The diagram shows intended reach relationships, not a certified ergonomic envelope. Seated and standing reach must be tested on Quest.

Object class	Visual language	Behaviour
Atoms and tools	Element symbols, clear silhouettes, hover feedback	Respond to grip/pinch; provide immediate state feedback
Fixed shelf/baskets	Matte tray surfaces, sockets and count labels	Stay fixed; only contents can be picked up
Background equipment	Muted materials; positioned beyond task reach	No hover response or grab interaction
Bond targets	Direction guides, arc and release ring	Appear only for valid authored opportunities

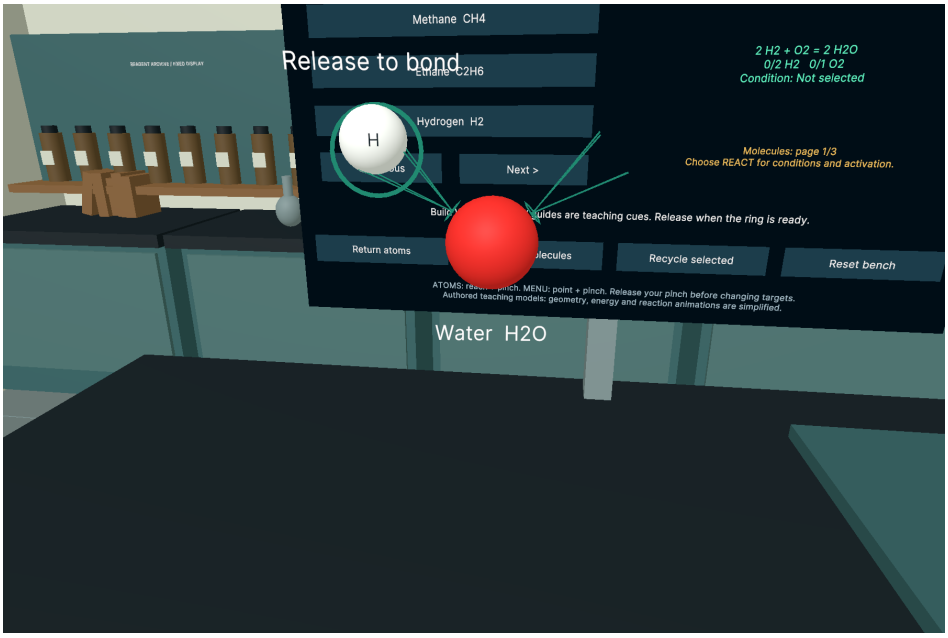
Avoid nearby nonfunctional handles or convincing grab cues. Physical placement, material treatment and behavioural consistency must reinforce each other; colour alone is insufficient. Background objects use inexpensive opaque materials, including simplified glassware.

Current visual implementation

These are actual Unity editor renders of the redesigned scene, captured on 28 September 2026. They illustrate implemented appearance and feedback, not headset performance or a completed user evaluation.



Room overview: the stock shelf is visually distinct from fixed background equipment.

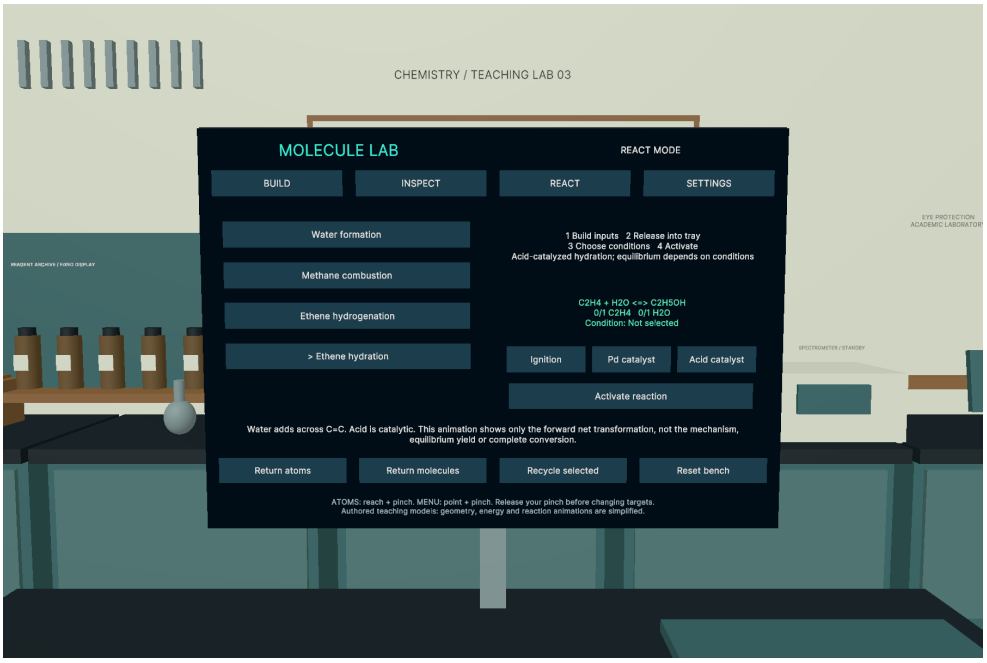


Staged valid placement: the direction guides and release ring show the pending bond.

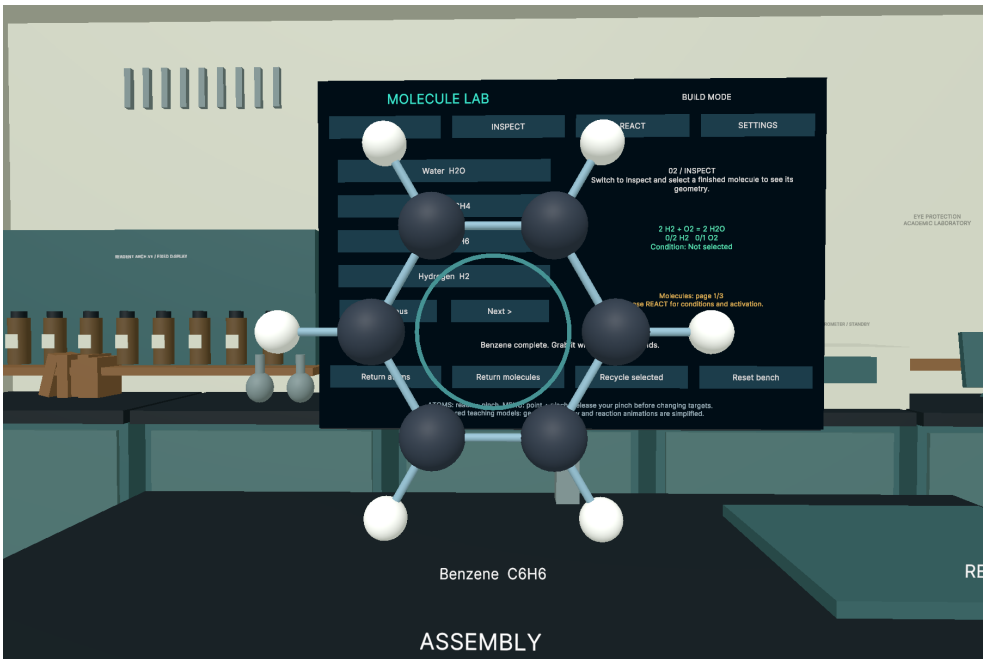
The second capture positions a hydrogen atom programmatically for visual inspection. It is not evidence of successful hand tracking. Device captures and participant observations remain separate evidence to collect.

Expanded chemistry and menu appearance

Actual Unity editor captures, 28 September 2026. The reaction panel is shown before input collection; the benzene model was assembled programmatically for visual verification. Neither capture demonstrates physical hand reliability or headset performance.



Reaction selection: inputs, conditions and an explicit activation step.



Benzene: planar ring, equivalent visible C–C bonds and delocalization circle.

Incomplete assembly stages are construction states, not proposed stable chemical intermediates. Correct completion uses authored geometry and valence constraints; no arbitrary chemistry prediction is implied.

Finite stock and recovery

Atoms and molecules are suspended, enlarged teaching models; this deliberate departure from real laboratory behaviour is consistent across tasks.

The shelf begins with **8 C, 8 O and 32 H**, all individually visible in four shallow baskets. Ordered sockets avoid a physics pile, obscure overlaps and uncontrolled bouncing. Empty sockets and counts make the system state visible. Basket arrangement is staggered front-to-back across two tiers in the reachable side rack.

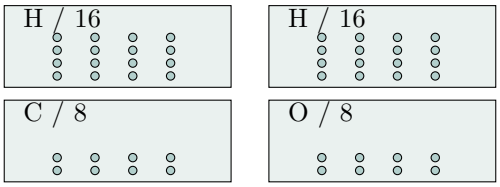


Figure 2. Four baskets, two tiers; every dot represents one atom.

The 32 H permit eight methane molecules or four ethanes with hydrogen left over. Oxygen supports water and the hydrogen/oxygen reaction. These capacities are alternatives, not a promise that all maximum quantities can coexist. Recycling enables continued practice.

Selecting a recipe moves a seed from the shelf into the assembly area. Switching an unfinished recipe returns its atoms. A completed molecule can be selected and recycled; recycling either radical returns both fragments, preventing an orphaned rejoin state. Loose atoms can be returned explicitly and are recovered if dropped outside the working area. Reset restores all 48 atoms and cancels outstanding animation.

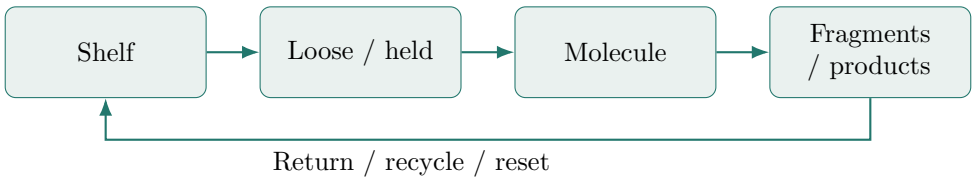


Figure 3. Ownership changes; total element counts remain constant.

Every token has a persistent identity and one owner. Reaction and fragment operations transfer identities rather than creating extra stock. During reaction animation, tokens remain owned by inputs until reassigned to products. No atom is silently discarded to satisfy an object-count limit.

Bonding: communicate before committing

Occupied carbon and oxygen show directional guides for valid neighbouring recipe sites. These are interface guides, not literal electron orbitals. A held compatible atom near a target produces a restrained electrical arc. A solid target ring and the words **Release to bond** confirm that releasing will attach it.

Initial tolerances are 18 cm for preview, 5 cm from the target for attachment, and 15 degrees from the authored bond direction. The direction is measured from the occupied atom toward the incoming atom, not from wrist orientation. These inspector settings are starting values for evaluation, not validated thresholds.

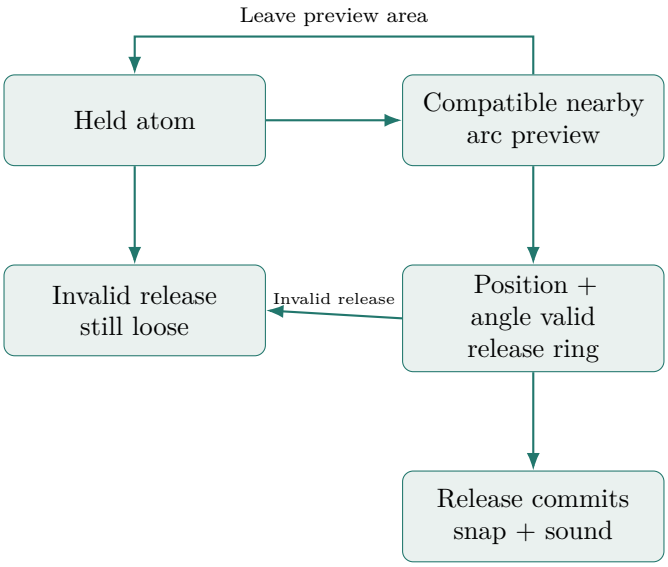


Figure 4. Bonding states; cancellation clears transient cues.

A shared evaluator decides both the preview and release outcome. A still-valid highlighted target remains stable if several targets are nearby. Release rechecks element, adjacency, vacant site, angle and distance; invalid attempts leave inventory and structure intact. Valid placement commits once, then animates a brief visual snap. Cancellation, reset, mode changes and tracking interruption clear active feedback.

Scientific interpretation. Covalent bonds are not electrical sparks. The tutorial explicitly identifies arcs as teaching cues. On completion, the transient effect disappears and solid rods retain the authored bond-order representation. The simulator remains recipe-based.

Evidence to collect. Observe at least one deliberately wrong-element release and one wrong-angle release. Record whether the learner can explain why each attempt failed, and whether the feedback predicts the next successful attempt.

Direct molecules, ray-operated menus

Implemented interaction design. Nearby thumb–index pinches acquire atoms and completed molecules. Hand rays operate menus only; fingertip menu presses and distant hand-ray object grabs are disabled. One pinch belongs to one target class until released. A hand holding an object cannot also operate a menu. Controllers retain grip grabs and ray/trigger selection.

The board has Build, Inspect, React and Settings pages, with four molecule choices per page. Buttons are at least 5 cm high with visible hover and pressed states. Height and forward/back controls support workstation reach. These choices address consistency, affordance, error prevention and recovery; effectiveness still requires headset observation.

Organic chemistry expansion

The added structures are ethene, ethyne, methanol, ethanol, carbon dioxide and benzene. Idealized sp, sp^2 and sp^3 geometry is distinguished from measured molecular dimensions. Benzene is planar with equivalent C–C bonds and a delocalization circle for six π electrons. Internal Kekulé bookkeeping is not presented as alternating physical bond lengths. Ring closure connects all required occupied neighbours once. Aromatic and multiple bonds do not use the ethane twist tool [5].

Oxygen's triplet ground state is identified; an O=O drawing alone is not a complete electronic model. Ethane homolysis yields two methyl radicals with an unpaired electron each. The energy tool is illustrative, not a measurement of bond dissociation energy.

Guided reaction selection

Net transformation	Condition represented
$2H_2 + O_2 \rightarrow 2H_2O$	Ignition
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Ignition; sufficient oxygen
$C_2H_4 + H_2 \rightarrow C_2H_6$	Pd-catalyzed hydrogenation
$C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$	Acid-catalyzed hydration

Select a reaction, build the listed inputs, dock them, select conditions, then activate. Live counts persist while building. Wrong quantities or conditions consume nothing; changing reactions undocks inputs. Catalysts are condition controls, not inventory atoms. Products remain reusable.

Hydrogenation requires its catalyst [4]. Hydration is reversible: the animation illustrates one forward net transformation, not equilibrium yield, guaranteed complete conversion or a mechanistic pathway [3]. Benzene, methanol and ethyne have build/inspect activities only in this update.

Evidence to collect. For each hand, record 20 direct grabs and 20 ray-menu selections; target at least 18 first-attempt successes per set and zero duplicate actions. Record actual chemistry review and headset findings, not inferred success from automated tests.

Animation and audio

Animation is reserved for state changes: seed transfer, bonding preview, a short snap, ethane torsion, reversible deformation and reaction rearrangement. There is no ambient object motion competing for attention. During reaction, the status text explains that the system controls the inputs; reset remains the escape route.

Quiet ventilation supplies room ambience. A brief ready tone, bond tone and completion tone are distinguishable but restrained. Sound can be muted; reduced effects remove arc motion while retaining the release state. Text and geometry convey every essential outcome without audio.

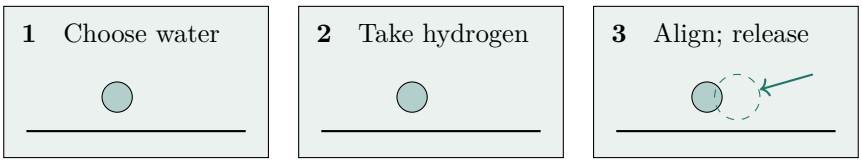


Figure 5. Interaction storyboard: context, acquisition, predicted outcome.

Assets and performance choices

Selected computer, keyboard, books, chair and plant meshes come from Kenney's CC0 Furniture Kit [2]. Their materials are replaced to match the room. Custom geometry supplies the shell, shelf, baskets, reagent bottles, glassware and instrument enclosure. The Free Laboratory Pack was researched as an optional source [6]; it is not required by the delivered scene. No third-party meshes are included in this report archive.

Use opaque materials, shared materials, simple static props and a bounded number of line effects. Avoid refractive glass, extra realtime effect lights and large transparent particle systems. Low file size and low polygon counts alone do not prove Quest performance.

Evidence to collect. Insert actual paper-prototype photographs here after testing the shelf and table layout. Capture the arrangement, task instructions and what changed as a result.

Paper prototype photograph — not yet collected

Evaluation protocol

Formative inspection. Seek 3–5 evaluators with HCI/VR knowledge where available. Inspect the environment against the course's Nielsen adaptations and VR-specific heuristics [1]. Report issue, task, heuristic, severity and proposed correction. Do this before a larger user study.

Novice pilot. Target five introductory learners for a small formative pilot, subject to course recruitment and consent requirements. This sample is for finding problems, not proving learning effectiveness. Record prior VR and chemistry experience; allow seated or standing use. Explain how to pause or stop and stop on discomfort.

Task	Evidence	Design rationale
Identify usable objects	First choices; decoration grab attempts	Affordance, recognition
Build water	Time to first bond; incorrect releases; help requests	Goals, feedback, error prevention
Build methane/ethane	Completion, hints used, perceived challenge	Scaffolding and flow
Recover an error	Return/recycle success; unintended loss	Control and freedom
Inspect and react	Geometry understanding; atom-count explanation	Domain compatibility
Switch inputs	Stuck selections; reach and tracking issues	Natural action, coordination

Record timestamps and observations on a worksheet or screen recording with participant agreement. The current software does not automatically collect research telemetry. Ask short post-task questions about clarity, control, perceived challenge, room credibility and comfort; label these as custom formative questions, not a validated flow scale.

Comparison. If the pilot reveals unclear bonding cues, compare two threshold/effect settings using the same tasks and counterbalanced order. Change one factor at a time. Keep results descriptive and report previous-task learning effects.

Device acceptance. Run the full lesson repeatedly and a 20-minute Quest 3 session. Record headset software, APK hash, CPU/GPU frame times, dropped frames, memory and thermal behaviour. Target sustained 72 FPS; include both hands and controllers, sleep/resume, reset during reaction and tracking interruptions.

ID	Observation / task	Severity	Heuristic / proposed fix
—	To be collected	—	—
—	To be collected	—	—

Evidence to collect. No participants have been tested for this redesign. Do not replace these blanks with inferred findings.

Implementation status and next steps

Implemented, 29 September 2026: direct hand grabbing with separate menu-only rays, paginated controls, shelf clearance adjustment, reach settings, 5 cm / 15 degree bond placement, eleven buildable molecules including aromatic benzene, and four condition-gated reactions. The finite C8/O8/H32 inventory remains unchanged.

Local verification: all 58 automated checks passed, covering chemistry, reaction conditions, inventory transfers, both-hand rig configuration, shelf clearance and recovery. These include programmatic scene walkthroughs, not physical hand trials. Build identification is recorded in BUILD-EVIDENCE.txt and the project implementation-status document.

Headset evidence: the user reported that the previous version worked, with inconsistent hand interaction. This update has not been physically validated on Windows Quest Link or standalone Quest. No new hand-success rates, user-study findings or performance results are claimed. The 20-minute / 72 FPS session remains pending.

Architecture and editable content

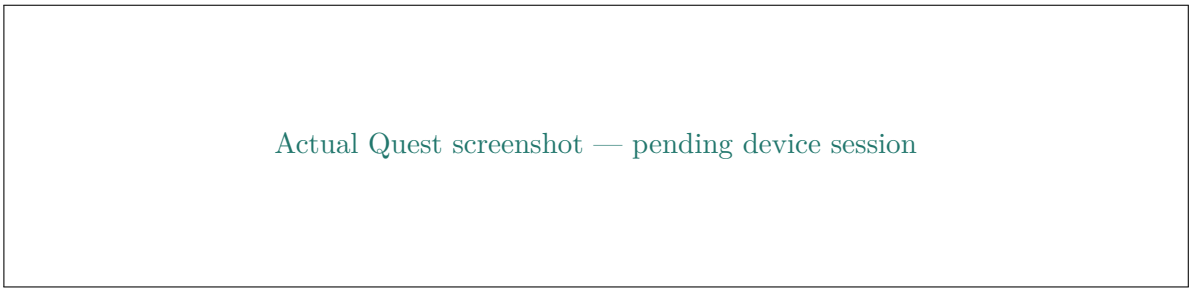
The academic environment is an editable prefab in the MoleculeLab Environment folder. Runtime initialization uses its bench, tray, panel and socket references. Chemistry definition generation preserves the existing scene. Decorators should edit the prefab or scene, not regenerate the environment to make small changes.

AtomInventory owns the finite token ledger. Loose atoms and molecule views reference identities; reactions and fragment operations transfer ownership. BondPlacement is the common pure placement evaluator; BondPreview presents its outcome. This separation supports testing rules independently of headset input.

The catalogue now includes H2O, CH4, C2H6, H2, O2, C2H4, C2H2, CH3OH, C2H5OH, CO2 and C6H6. A new molecule requires authored geometry and bonds, an inventory feasibility check and interaction testing. Arbitrary structure generation remains outside this treatment.

Outstanding design evidence

- Reach all basket sockets while seated and standing without leaning through the table.
- Verify that fixed room detail is understood as scenery and does not distract from atoms.
- Tune cue thresholds after observing wrong-angle and wrong-element attempts.
- Check text, muted audio, reduced effects and visibility with both hands and controllers.
- Confirm balanced chemistry labels with an appropriate course reviewer.



References

[1] James Gain. Virtual reality: Evaluating virtual environments. University of Cape Town, Department of Computer Science, course slides supplied by the project owner. 17 slides; local file VR-2-Evaluation.pdf. Publication date not specified.

[2] Kenney. Furniture kit. https://kenney.nl/assets/furniture-kit. CC0 asset pack. Accessed 25 September 2026.

[3] John McMurry. Organic chemistry: Preparation of alcohols: A review. https://openstax.org/books/organic-chemistry/pages/17-3-preparation-of-alcohols-a-review, 2023.

[4] John McMurry. Organic chemistry: Reduction of alkenes: Hydrogenation. https://openstax.org/books/organic-chemistry/pages/8-6-reduction-of-alkenes-hydrogenation, 2023.

[5] John McMurry. Organic chemistry: Structure and stability of benzene. https://openstax.org/books/organic-chemistry/pages/15-2-structure-and-stability-of-benzene, 2023.

[6] StormBringer Studios. Free laboratory pack. https://assetstore.unity.com/packages/3d/props/tools/free-laboratory-pack-123782. Research candidate, not imported. Accessed 25 September 2026.